

EE 110: Circuits Analysis and Design

(3rd Semester, Fall 2022)

Instructors

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Week	Experiment	Topics
1	---	Introduction to Laboratory Equipment
2	1	To examine the pulse response of a series RL network
3	2(a)	To study the steady state sinusoidal response of RL and phasors
4	2(a)	To study the steady state sinusoidal response of RC and phasors
5	2(b)	To show the resonant frequency of a series RLC circuit is given by $1/(2\pi\sqrt{LC})$ and to plot the frequency response of an RLC circuit
6	3	To plot the magnitude and phase response of passive low pass and high pass RC filters
MID-TERM EXAMS		
8	4	To study the frequency characteristics of passive low pass and high pass RL filter circuits
9	5	To plot the magnitude and phase response of a series resonant band-pass filter
10	6	To plot the magnitude and phase response of a series resonant band-stop filter
11	7	Implementation of first order Active low pass and high pass filter
12	8	To plot the magnitude and phase response of a Active Band Pass and Band Stop filters
13	Project	Second order active low and high pass filter
14		Second order active band pass and band stop filter
Final-Term Exams		

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Experiment 1: To examine the pulse response of a series RL network.

In this experiment, we apply a pulse waveform to the RL circuit to analyze the transient response of the circuit. The pulse-width relative to the circuit's time constant determines how it is affected by the RL circuit.

Rise of current

At the instant when step voltage is applied to an RL network, the current increases gradually and takes some time to reach the final value. The reason the current does not build up instantly to its final value is that as the current increases, the self-induced e.m.f. in L opposes the change in current (Lenz's Law). Mathematically, it can be shown,

$$i(t) = \frac{V}{R} (1 - e^{-t/\tau})$$

Where t = time elapsed since pulse is applied
 $\tau = L/R$ = Time constant of the circuit

Decay of current

During the next half cycle of pulse, when the pulse amplitude is zero, the current decreases to zero exponentially. Mathematically, it can be shown,

$$i(t) = \frac{V}{R} e^{-t/\tau}$$

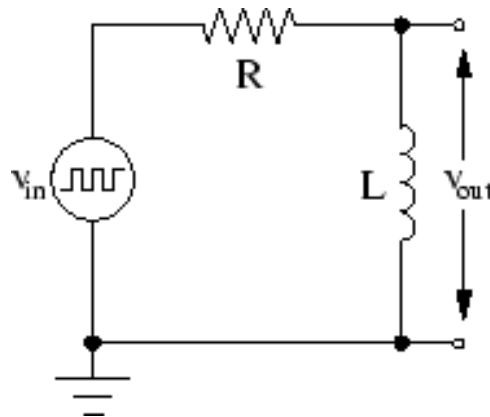
Equipment and Components required:

- Oscilloscope
- Signal Generator
- Inductors
- Resistors(10k Ω /20k Ω /Use resistor of standard value available in the Lab)

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Circuit to be patched on breadboard:



Laboratory Tasks:

1. Set the output of the function generator to a square-wave with peak-to-peak amplitude of 5V and adjust the frequency that should allow the inductor to charge and discharge fully i.e. **f should be lesser than or equal to $f_{max} = 1/10\tau$** . Use dc offset to obtain pulse train ranging from 0 to 5 volts from square wave.
2. Patch the given circuit on the breadboard (Use appropriate values of R and L so that you can easily measure the voltage levels from the oscilloscope at different values of τ) and apply pulse train to it.
3. Display simultaneously voltage $V_{in}(t)$ across the function generator (**on CH 1**) and $V_L(t)$ across the inductor L (**on CH 2**). Ensure common ground for both channels and the signal generator.
4. Sketch the two measured waveforms $V_{in}(t)$ and $V_L(t)$, calculate and sketch the waveforms, $V_R(t)$ and $i(t)$. Label the time, voltage and current scales. Note that the voltage across resistor R , $V_R(t)$, also represents the current $i(t)$.
5. Measure the time constant, τ , using the waveform $V_L(t)$. Expand the time scale and measure the time it takes for the waveform to complete 63% of its total change, i.e. 5V. Enter the measured value of τ in Table.
6. Compare values of theoretically expected and experimentally obtained time constants τ .
7. Now calculate and measure $i(t)$ for five different values of τ for rise of current as well as fall of current.

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Observations & Calculations:

No.	R	L	T	10τ	$f_{\max}$
1	1K Ω	10mH	10 μ s	100 μ s	10KHz

Rise of current:

Calculated current: $i(t) = \frac{V}{R} (1 - e^{-t/\tau})$

Measured Current from voltage: From Oscilloscope

No. of Time Constants	Calculated Current (mA)	Measured Current (mA)
τ	3.16	2.9
2τ	4.32	3.8
3τ	4.75	4.5
4τ	4.91	4.6
5τ	4.96	4.7

Decay of current:

Calculated current: $i(t) = \frac{V}{R} e^{-t/\tau}$

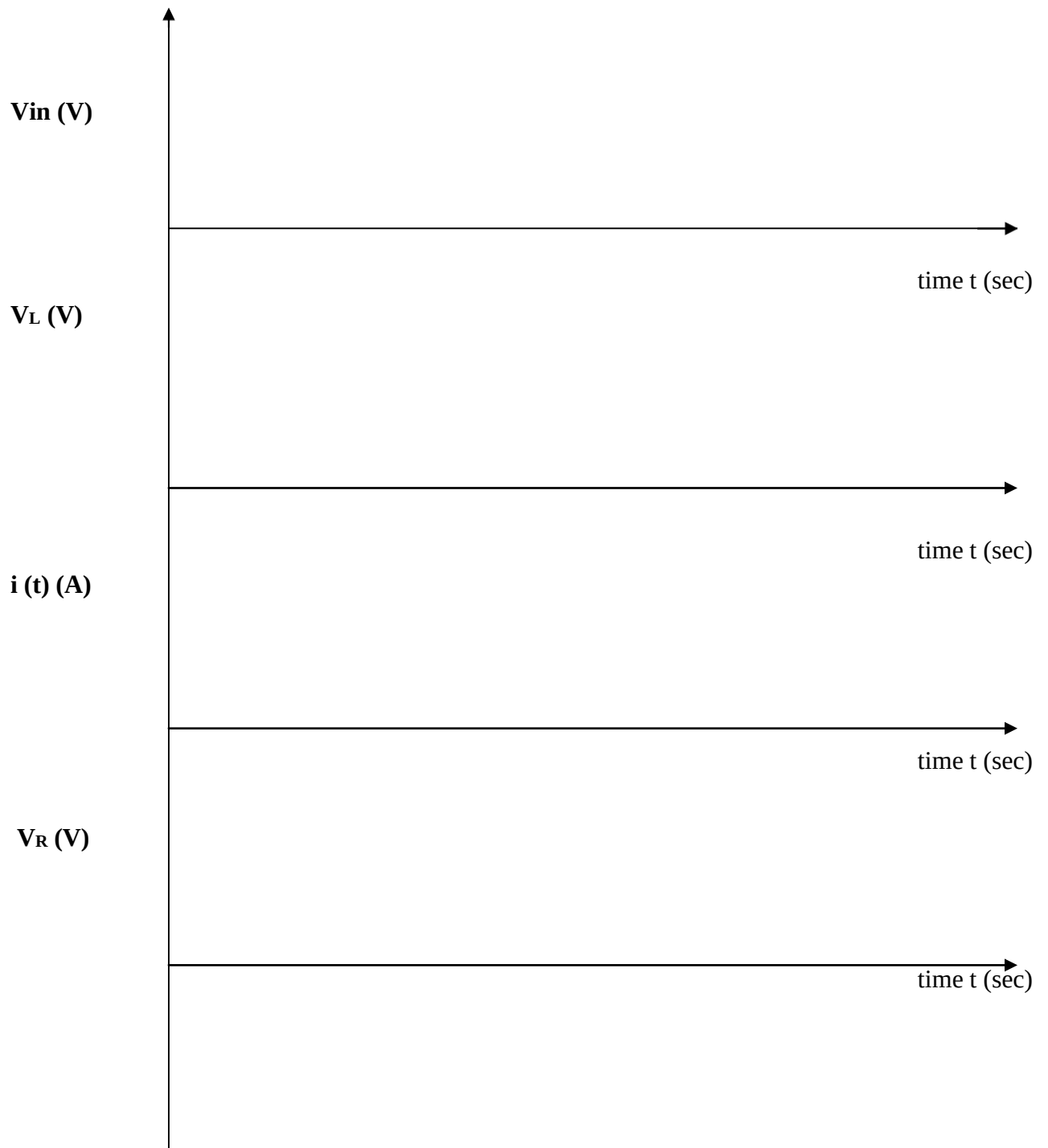
Measured Current from voltage: From Oscilloscope

No. of Time Constants	Calculated Current (mA)	Measured Current (mA)
τ	1.84	1.6
2τ	0.67	0.5
3τ	0.25	0.2
4τ	0.09	0.05
5τ	0.03	0

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WAVEFORMS OF VOLTAGES & CURRENT



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Laboratory Report:

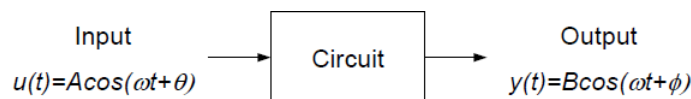
1. Briefly explain the behavior exhibited by V_L and V_R waveforms.
2. Connect a resistor of the same value in parallel with already connected resistor and observe the change in the waveforms of V_L and V_R ?
3. Connect a inductor of the same value in parallel with already connected capacitor and observe the change in the waveforms V_L and V_R ?
4. Plot V_L and V_R waveform for each case and compare them.

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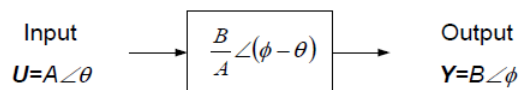
Experiment 2: (A) To study the steady state sinusoidal response (RC and RL circuits) and phasors.

(B) To show the resonant frequency of a series RLC circuit is given by $1/(2\pi\sqrt{LC})$ and to plot the frequency response of an RLC circuit

In this lab we will be concerned with the steady-state response of electrical circuits to sinusoidal inputs. Figure 1(a) shows a block-diagram representation of the system. The input is a cosine function with amplitude A and phase angle θ . The output is a cosine function with amplitude B and phase angle ϕ . Both the input and output waveforms have radian frequency ω (recall that an important property of linear systems is that the steady-state response of a linear system to a sinusoidal input is a sinusoid with the same frequency as the input sinusoid). The analysis of the circuit of Figure 1(a) can be simplified by representing the sinusoidal signals as phasors. The phasors provide the amplitude and phase information of the sinusoidal input and output signals. The input-output relationship governing the circuit then reduces to a relationship between the output and input signal amplitudes and the output and input signal phases. The circuit can thus be represented in phasor form as an amplitude gain between the output and input signals and a phase difference between the output and input signals, as shown in Figure 1(b).



(a) Physical circuit



(b) Phasor representation of circuit input-output relationship.

Figure 1. Steady-state sinusoidal circuit analysis

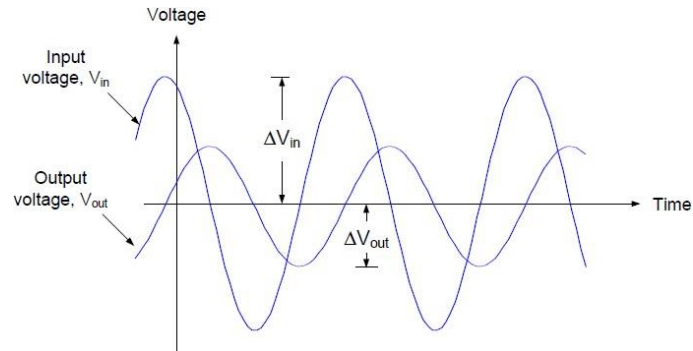
The gain of a system at a particular frequency is the ratio of the magnitude of the output voltage to the magnitude of the input voltage at that frequency, so that:

$$\text{Gain} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

where ΔV_{out} and ΔV_{in} can be measured from the sinusoidal input and output voltages as shown in the figure,

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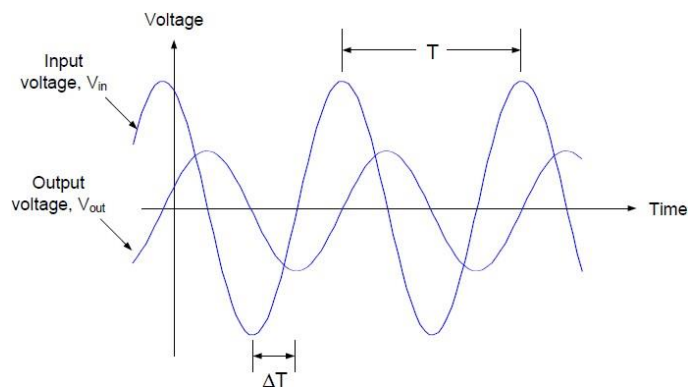
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The phase of a system at a particular frequency is a measure of the time shift between the output and input voltage at that frequency, so that:

$$\text{Phase} = \frac{\Delta T}{T} \times 360^\circ$$

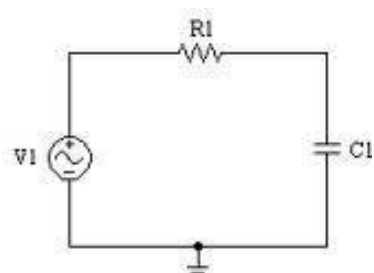
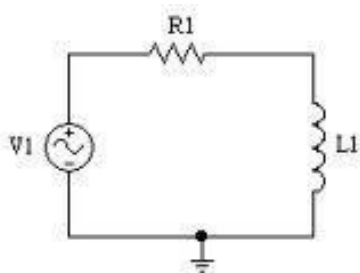
where ΔT and T can be measured from the sinusoidal input and output voltages as shown in the figure below,



Equipment and Components required:

- Oscilloscope
- Signal Generator
- Inductors and Capacitors
- Resistors(Use resistor of standard value available in the Lab)

Circuits to be patched on breadboard:



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Sinusoidal Response of RL circuit:

Circuits containing inductance and resistance appear in many electronic circuits ranging from power supplies to filters. In this experiment we have to investigate the sinusoidal response of series RL circuit. A difficulty that arises in such circuits is that real inductors are not like the ideal inductors. A real inductor is formed of coiled wire so it possesses resistance as well as inductance. Furthermore for some inductors the resistance is dependent on frequency as well. As a consequence, the resistance R inserted in series with the Inductor does not represent the total resistance of the circuit. In this lab task we will try to overcome this problem by making R large as compared to resistance of the Inductor coil i.e. we will assume the coil to be an ideal one.

$$Z_L = X_L = j\omega L \text{ (assuming resistance of Coil to be zero)}$$

$$Z_{TOTAL} = \sqrt{R^2 + X_L^2}$$

$$I = V / Z_{TOTAL} \text{ (V and } Z_{TOTAL} \text{ are complex terms)}$$

$$V_R = IR$$

$$V_L = IZ_L$$

Laboratory Tasks:

1. Calculate and note down quantities Z_L , Z_{TOTAL} , I , V_R and V_L for a source voltage of 5V peak and a frequency of 1kHz.
2. Patch the given circuit on the breadboard (Use appropriate values of R and L) and apply sinusoidal signal from the signal generator. Adjust the magnitude and frequency to the values chosen above.
3. Measure voltage V_L (magnitude and phase with respect to input) and V_R (magnitude and phase with respect to input) and compare it with the calculated values.
4. Explain any difference between measured and calculated values of V_L and V_R .
5. Draw the phasor diagram of the measured voltages. Include I as a reference phasor and show positions of V , V_L and V_R .
6. Find the frequency when the phase difference between V (input voltage) and V_R becomes 45 degrees. What is the Gain at this frequency?

Observations & Calculations:

Parameter	Freq	Z_L	Z_{TOTAL}	I (mA)	V_R	V_L (V/s)
Calculated	50	3.14	10K	0.5	5	0.00157
	500	31.4	10K	0.5	5	0.0157

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	5K	314	9995	0.5	5	0.157
	50K	3140	9494	0.5	5	1.57

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frequency	$Z_L = V_L/I$	$Z_{TOTAL} = V/I$	$I = V_R/R$	V_R	V_L	$\emptyset$
50	76	10156	0.5	5.04	0.038	20
500	96	10016	0.5	4.96	0.048	39.04
5K	364	10289	0.5	4.96	0.182	42
50K	3440	10200	0.328	3.38	1.72	66

Frequency: 60K (When input voltage and V_R have a 45° phase difference).

Gain at this frequency: 0.96

Sinusoidal Response of RC circuit:

$Z_C = X_C = 1/j\omega C$ (assuming resistance of Coil to be zero)

$Z_{TOTAL} = \sqrt{R^2 + X_C^2}$

$I = V / Z_{TOTAL}$ (V and Z_{TOTAL} are complex terms)

$V_R = IR$

$V_C = IZ_C$

Laboratory Tasks:

1. Calculate and note down quantities Z_C , Z_{TOTAL} , I , V_R and V_C for a source voltage of 5V peak and a frequency of 1kHz.
2. Patch the given circuit on the breadboard (Use appropriate values of R and C) and apply sinusoidal signal from the signal generator. Adjust the magnitude and frequency to the values chosen above.
3. Measure voltage V_C (magnitude and phase with respect to input) and V_R (magnitude and phase with respect to input) and compare it with the calculated values.
4. Explain any difference between measured and calculated values of V_C and V_R .
5. Draw the phasor diagram of the measured voltages. Include I as a reference phasor and show positions of V, V_C and V_R .
6. Find the frequency when the phase difference between V (input voltage) and V_R becomes 45 degrees. What is the Gain at this frequency?

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Observations & Calculations:

Parameter	Z_C	Z_{TOTAL}	I	V_R	V_C
Calculated	3.38	1000	0.005	5	0.0169

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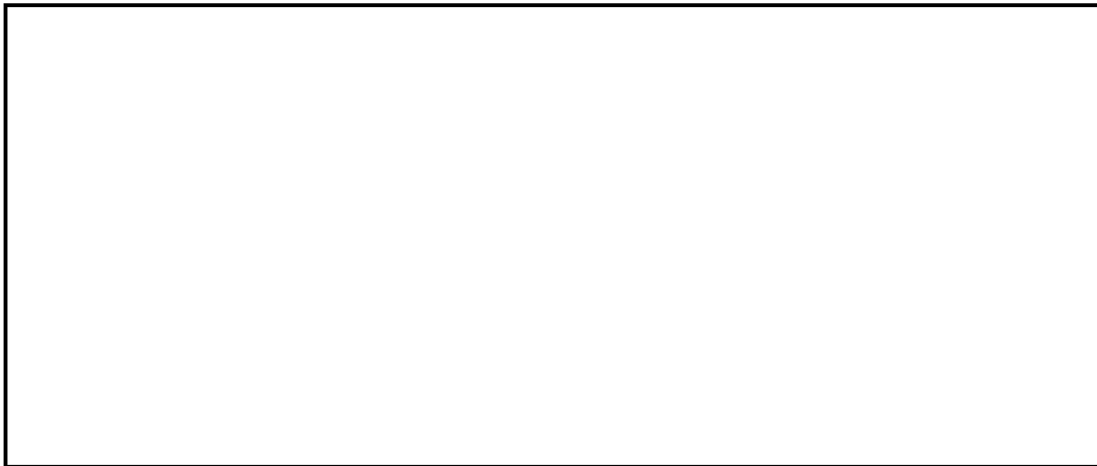
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Parameter	$Z_C = V_C/I$	$Z_{TOTAL} = V/I$	$I = V_R/R$	V_R	V_C
Measured					

Frequency: _____ (When input voltage and V_R have a 45° phase difference).

Gain at this frequency: _____.

Phasor Diagrams:

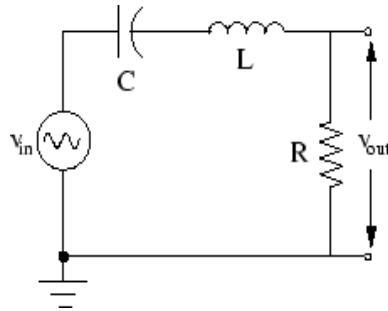


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PART (B): To show the resonant frequency of a series RLC circuit is given by $1/(2\pi\sqrt{LC})$ and to plot the frequency response of an RLC circuit

Circuit to be patched on breadboard:

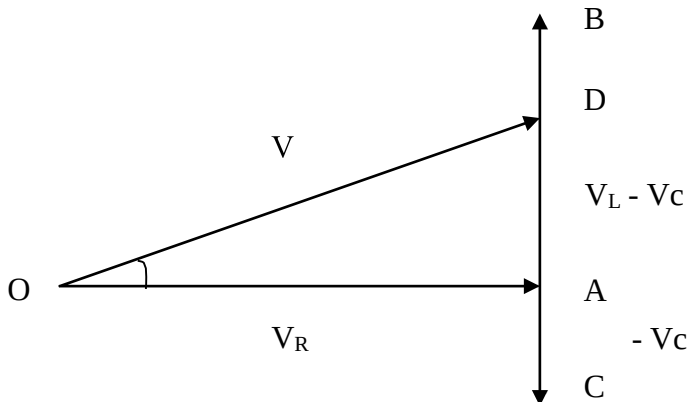


As shown in the circuit diagram, resistor, inductor and capacitor are connected in series with an a.c. supply of r.m.s. voltage V . The phasor diagram is plotted as,

$V_R = IR$ = voltage drop across R

$V_L = IX_L$ = voltage drop across L

$V_C = IX_C$ = voltage drop across C



In voltage triangle of figure, OA represents V_R , AB and AC represents the inductive and capacitive drops respectively. It will be seen that V_L and V_C are 180 degree out of phase with each other i.e. they are in direct opposition to each other. Subtracting AC from AB , we get the net reactive drop $AD = I(X_L - X_C)$. The applied voltage V is represented by OD and is the vector sum of OA and AD .

$$OD = \sqrt{OA^2 + AD^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2} = I\sqrt{(R)^2 + (X_L - X_C)^2}$$

$$I = V / \sqrt{(R)^2 + (X_L - X_C)^2} = V/Z$$

The term $Z = \sqrt{(R)^2 + (X_L - X_C)^2}$ is known as the **impedance** of the network.

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Resonance in RLC Networks:

Resonance means to be in step with. When the applied voltage and the current in an a.c. network are in step with i.e. phase angle between voltage and current is zero or $\text{pf} = 1$, the circuit is said to be in resonance. An a.c. circuit containing reactive element (L and C) is said to be in resonance when the net reactance is zero.

When a series R-L-C is in resonance, it possesses minimum impedance $Z = R$. Hence, circuit current is maximum, it being limited by value of R alone. The current $I_0 = V/R$ and is in phase with V. since circuit current is maximum, it produces large voltage drops across L and C. but these drops being equal and opposite, cancel out each other. Taken together, L and C form part of a circuit across which no voltage develops however, large the current flowing. The frequency at which the net reactance of the series circuit is zero is called the resonant frequency. Its value can be found as under:

$$\begin{aligned} X_L - X_C &= 0 \\ \text{OR } X_L &= X_C \text{ or } \omega_0 L = 1/\omega_0 C \\ \text{OR } \omega_0^2 &= 1/LC \text{ or } (2\pi f_0)^2 = 1/LC \text{ or } f_0 = 1/2\pi \sqrt{LC} \end{aligned}$$

If L is in Henry and C is in Farad, then f_0 is in Hertz.

Laboratory Tasks

1. For the given inductor and capacitor calculate the resonant frequency and connect the circuit as shown in circuit diagram
2. Apply sinusoidal signal from the signal generator of 5V_{peak} to the network and set the frequency to a value of 500 Hz
3. Observe V_R on the oscilloscope and record it.
4. Increase the frequency of the signal and for each frequency measure and record V, V_R , V_L and V_C and maintain applied voltage constant at 5V_{pk}
5. Now measure V_R , V_L and V_C theoretically and compare the results.

Observations & Calculations:**Measured Values**

No.	Frequency f (Hz)	$X_L = 2\pi fL$ (ohms)	$X_C = 1/2\pi fC$ (ohms)	$Z = \sqrt{(R)^2 + (X_L - X_C)^2}$ (ohms)	$I = V_R/R$ (Amp)	$V_L = IX_L$ (volts)	$V_C = IX_C$ (volts)	V_R
1	20	1.256	795.77	794.51	0.0036	0.005	0.0038	3.6
2	50	3.14	318.3	10K	0.0005	0.002	0.160	4.8
3	503	31.60	31.64	10004	0.0005	0.0176	0.0168	4.8
4	1K	62.83	15.91	10K	0.0005	0.032	0.0096	4.8
5	50K	31415.9	0.0318	31431.8	0.00152	4.8	0.0044	1.52

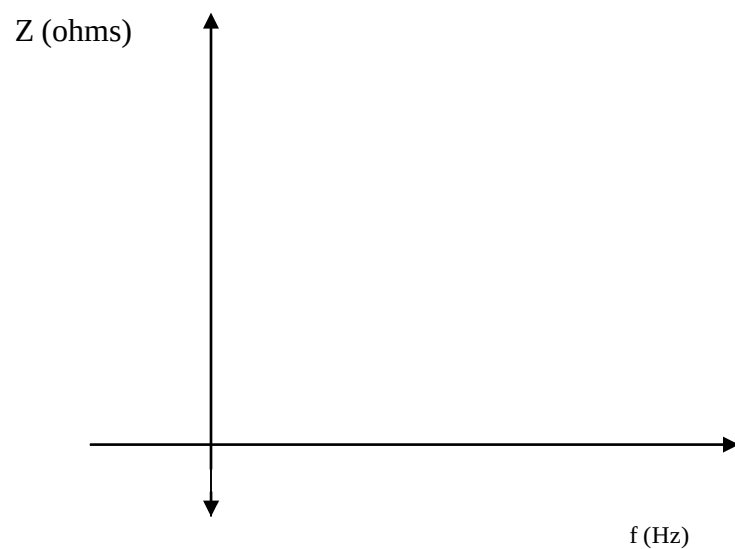
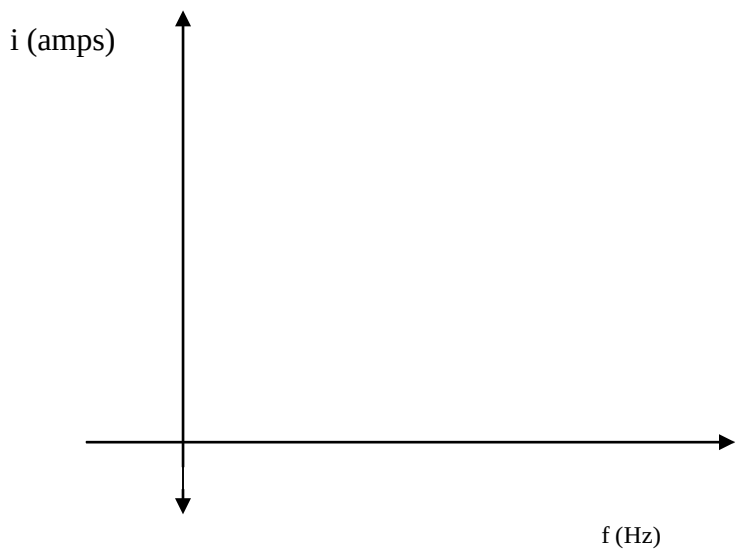
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Calculated Values

No.	Frequency f (Hz)	$X_L = 2\pi fL$ (ohms)	$X_C = 1/2\pi fC$ (ohms)	$Z = \sqrt{R^2 + (X_L - X_C)^2}$ (ohms)	$I = V_R/R$ (Amp)	$V_L = IX_L$ (volts)	$V_C = IX_C$ (volts)

GRAPH



Laboratory Report:

PART (A)

1. Briefly explain the phasor diagrams drawn for both the circuits.
2. Connect a resistor of the same value in parallel with already connected resistor for both the circuits and observe the change (magnitude and phase) in V_L and V_R ?
3. Connect a capacitor of the same value in parallel with already connected capacitor and observe the change (magnitude and phase) in V_C and V_R ?

PART(B)

4. Explain the waveforms plotted in the graph.
5. Comment on the effect of increase in frequency on X_C , X_L and I .

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Experiment 3: To plot the magnitude and phase response of passive low pass and high pass RC filters.

By using various combinations of resistance's, capacitors and inductors we can make circuits that have the property of passing or rejecting either low or high frequencies or bands or frequencies. These frequency selective networks, which alter the amplitude and phase characteristics of the input ac signal, are called filters. Electronic filters can be passive or active. Passive implementations of filters are based on combinations of resistors (R), inductors (L) and capacitors (C). These types are collectively known as *passive filters*, because they do not depend upon an external power supply and/or they do not contain active components such as transistors. Active filters are implemented using a combination of passive and active (amplifying) components, and require an outside power source. Operational amplifiers are frequently used in active filter designs. However, their upper frequency limit is limited by the bandwidth of the amplifiers used.

IMPLEMENTATION OF LOW PASS FILTER:

Low pass filter: A low pass filter allows signals with lower frequencies to pass from input to output while rejecting higher frequencies.

Pass band of a filter: The range of low frequencies passed by a low-pass filter within a specified limit is called the pass band of the filter.

Critical Frequency: The critical frequency is the frequency at which the filter's output voltage is 70.7% of the maximum. The filter's critical frequency is also called the *cutoff frequency*, *break frequency*, or *-3 dB frequency* because the output voltage is down 3 dB from its maximum at this frequency. The term *dB (decibel)* is a commonly used one that you should understand because the decibel unit is used in filter measurements.

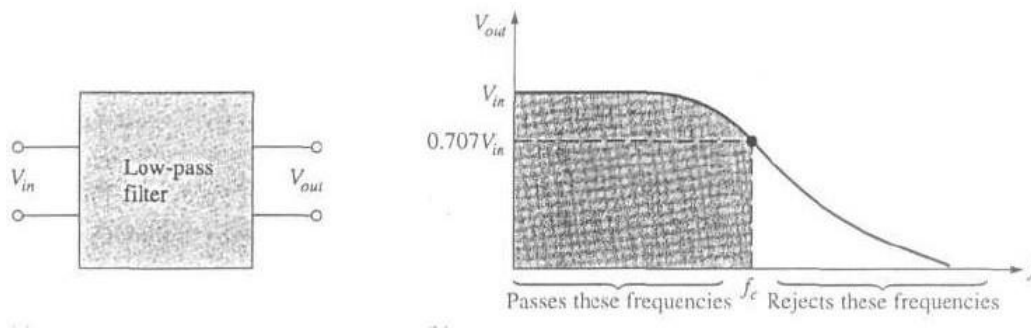


FIGURE: Low-pass filter block diagram and general response curve.

Decibels

The basis for the decibel unit stems from the logarithmic response of the human ear to the intensity of sound. The decibel is a logarithmic measurement of the ratio of one power to another or one voltage to another, which can be used to express the input-to-output relationship of a filter. The following equation expresses a voltage ratio in decibels:

$$\text{db} = 20 \log (V_o / V_{in})$$

Operation of the LPF

When the input is dc (0 Hz), the output voltage equals the input voltage because X_c is infinitely large. As the input frequency is increased, X_c decreases and, as a result, V_{out} , gradually decreases until a frequency is reached where $X_c = R$. This is the critical frequency, f_c , of the filter.

$$X_c = R$$

$$\frac{1}{2\pi f_c C} = R$$

$$f_c = \frac{1}{2\pi RC}$$

At the critical frequency, the output voltage magnitude is

$$V_{out} = \left(\frac{X_c}{\sqrt{R^2 + X_c^2}} \right) V_{in}$$

by application of the voltage-divider formula. Since $X_c = R$ at f_c , the output voltage can be expressed as

$$\begin{aligned} V_{out} &= \left(\frac{R}{\sqrt{R^2 + R^2}} \right) V_{in} = \left(\frac{R}{\sqrt{2R^2}} \right) V_{in} \\ &= \left(\frac{R}{R\sqrt{2}} \right) V_{in} = \left(\frac{1}{\sqrt{2}} \right) V_{in} = 0.707 V_{in} \end{aligned}$$

These calculations show that the output is 70.7% of the input when $X_c = R$. The frequency at which this occurs is, by definition, the critical frequency.

The ratio of output voltage to input voltage at the critical frequency can be expressed in decibels as follows.

$$V_{out} = 0.707V_{in}$$

$$\frac{V_{out}}{V_{in}} = 0.707$$

$$20 \log\left(\frac{V_{out}}{V_{in}}\right) = 20 \log(0.707) = -3 \text{ dB}$$

Ideal and Actual Response of LPF

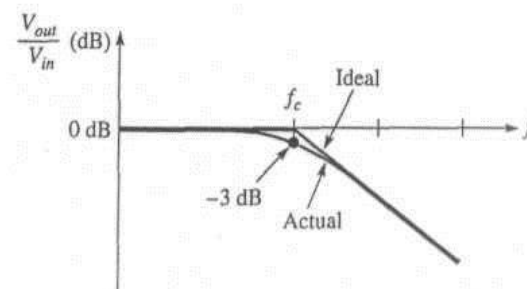


FIGURE: Actual and ideal response curves for a low-pass filter

Phase Shift in a Low-Pass Filter

The RC low-pass filter acts as a lag network. Since that the phase shift from input to output is expressed as

$$\phi = -90^\circ + \tan^{-1}\left(\frac{X_C}{R}\right)$$

At the critical frequency, $X_C = R$ and, therefore, $\phi = -45^\circ$. As the input frequency is reduced, ϕ decreases and approaches 0° as the frequency approaches zero.

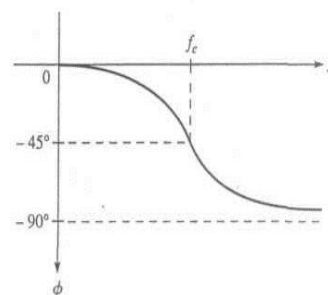
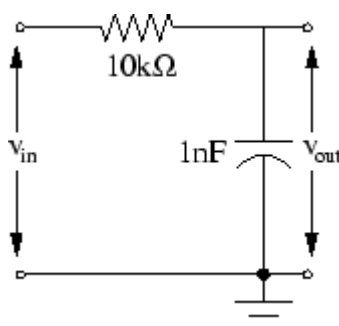


FIGURE: Phase characteristic of a Low-pass filter

Equipment and Components required:

- Oscilloscope
- Signal Generator
- Capacitors (0.1 μ F, 1nF)
- Resistors(1K, 10K, Use resistor of standard value available in the Lab)

Circuit to be patched on breadboard:**Laboratory Tasks:**

1. Apply a 1 Vpp 100 Hz signal as input to the network of LPF and measure the corresponding output voltage level. Determine the decibel gain of the filter.

$$G \text{ (dB)} = 20 \log [V_o/V_{in}]$$
2. Determine the phase difference between V_o and V_{in} in degrees.
3. Repeat step 1 and 2 for the following frequencies: 200 Hz, 500Hz, 1 KHz, 2 KHz, 5 KHz, 10 KHz, 20 KHz, 50 KHz.

Observations & Calculations:

Cutoff Frequency: $f_c = 1/2\pi RC = 15.9 \text{ Hz}$

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	db = 20 log (V_o/V_{in})	$\phi = -90 + \tan^{-1}(X_c/R)$ (degrees)
1	500	1.98	1.98	1	0	-1.8
2	1K	1.98	1.98	1	0	-3.6
3	10K	1.98	1.64	0.83	-1.64	-32.14
4	15.9K	1.98	1.34	0.68	-4.87	-45
5	20K	1.98	1.18	0.60	-4.50	-51.5
6	50K	1.98	0.58	0.30	-10.66	-72.34
7						
8						
9						

IMPLEMENTATION OF HIGH PASS FILTER:

High Pass Filter: A high pass filter allows signals with higher frequencies to pass from input to output while rejecting lower frequencies.

Critical Frequency: The frequency considered to be lower end of pass-band is called the critical frequency. It is the frequency at which the output is 70.7% of the maximum. In RC high pass filter output is taken across the resistor.

Lower frequencies experience considerable reactance by the capacitor and are not easily passed. Higher frequencies encounter a little reactance and are easily passed. The high frequencies passing through the filter develop the output voltage V_o across R , all frequencies above f_c are passed where as those below it are attenuated. At f_c , $R = X_C$ and the phase angle between V_o and V_{in} is 45° . For an ideal high pass filter the output is like this.

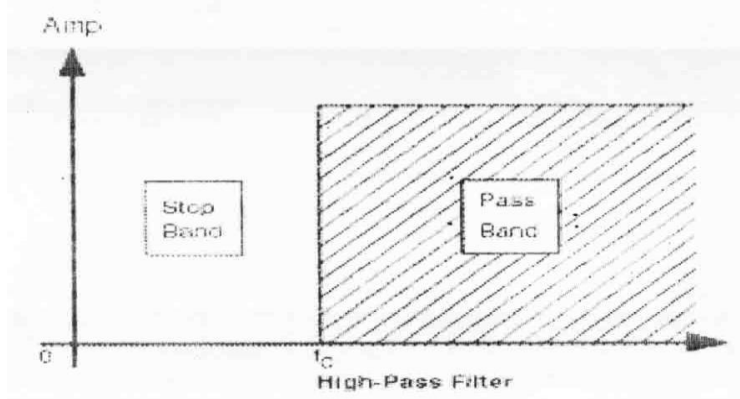


FIGURE: Response of an ideal High Pass filter

Bode Plot

When the input frequency is at its critical value, $X_C = R$ and the output voltage is $0.707V_{in}$, just as in the case of the low-pass filter. As the input frequency increases above f_c , X_C decreases and, as a result, the output voltage increases and approaches a value equal to V_{in} . The expression for the critical frequency of the high-pass filter is the same as for the low-pass filter.

$$f_c = \frac{1}{2\pi RC}$$

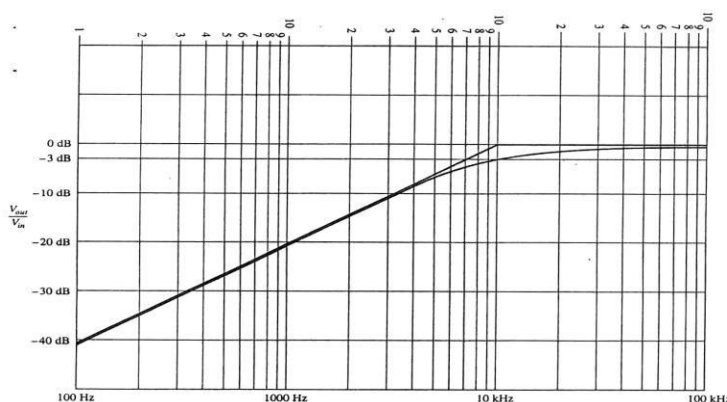


FIGURE: Bode plot for High Pass Filter. Upper line is the ideal and lower line is the actual.

Phase Shift in a High-Pass Filter

Both the RC and the RL high-pass filters act as lead networks. Recall that the phase shift from input to output for the RC lead network is:

$$\phi = \tan^{-1} \left(\frac{X_C}{R} \right)$$

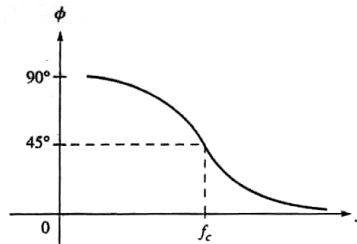
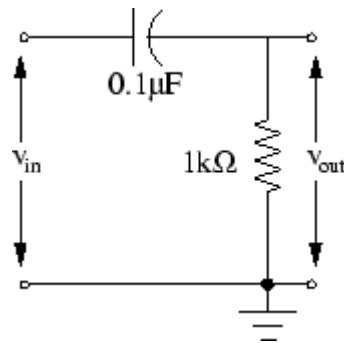


FIGURE: Phase characteristic of a high-pass filter.

Circuit to be patched on breadboard:



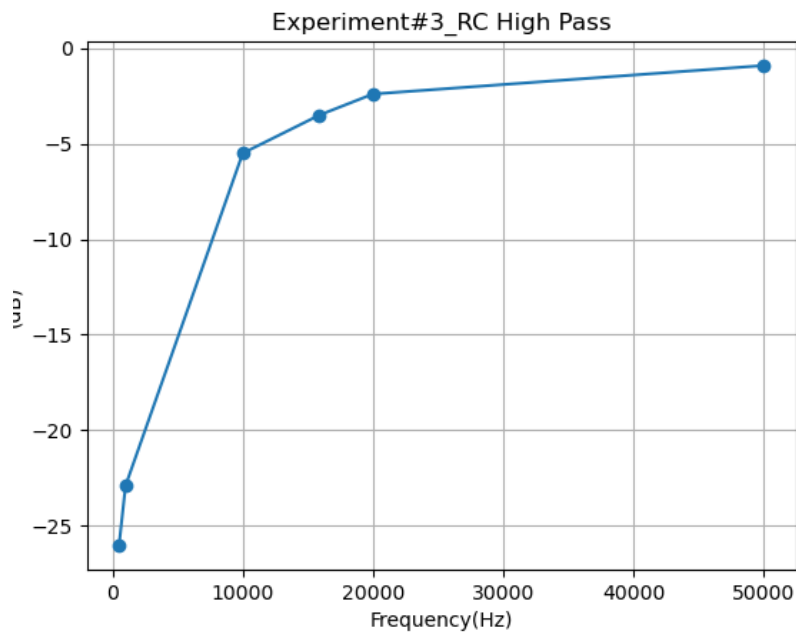
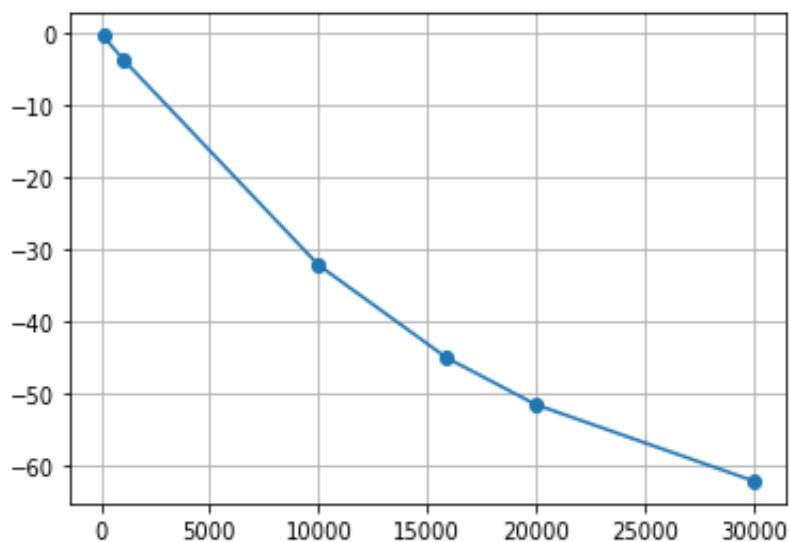
Observations & Calculations:

Cutoff Frequency: $f_c = 1/2\pi RC$

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	$db = 20 \log (V_o/V_{in})$	$\phi = \tan^{-1}(X_c/R)$ (degrees)
1	500	1.98	0.1	0.05	-26.02	88.2
2	1K	1.98	0.16	0.08	-22.9	86.4
3	10K	1.98	1.06	0.53	-5.5	57.86
4	15.9K	1.98	1.38	0.67	-3.50	45
5	20K	1.98	1.52	0.76	-2.4	37.5
6	50K	1.98	1.80	0.90	-0.91	17.66

Laboratory Report:

1. Plot the graph between gain in db against frequency and Phase angle against frequency for low pass filter?
2. Plot the graph between gain in db against frequency and Phase angle against frequency for high pass filter?

**RC Low Pass:**

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Experiment 4: To study the frequency characteristics of passivelow pass and high pass RL filter circuits.

By using various combinations of resistances, capacitors and inductors we can make circuits that have the property of passing or rejecting either low or high frequencies or bands of frequencies. These frequency selective networks which alter the magnitude and phase characteristics of the input ac signal are called filters. So we can say that filter is an AC circuit that separates some frequencies from others within mixed frequency signals. In the previous lab we implemented low pass and high pass RC filters. In this lab we have to implement RL low pass and high pass filters.

IMPLEMENTATION OF LOW PASS FILTER:

As we know that a low pass filter allows signals with lower frequencies to pass from input to output while rejecting higher frequencies. So in order to implement low pass RL filter the output voltage has to be taken across resistor. When the input is DC (0Hz) the output voltage is ideally equal to input voltage because X_L is a short circuit. As the input frequency is increases X_L increases and as a result V_{OUT} gradually decreases and it becomes $0.707V_{IN}$ at critical frequency.

Phase Shift in a Low-Pass Filter

The RL low pass filter act as a Lag network. At critical frequency when $X_L=R$ phase between output and input is -45° . The phase difference from input to output varies from 0° to -90° as the frequency is increased.

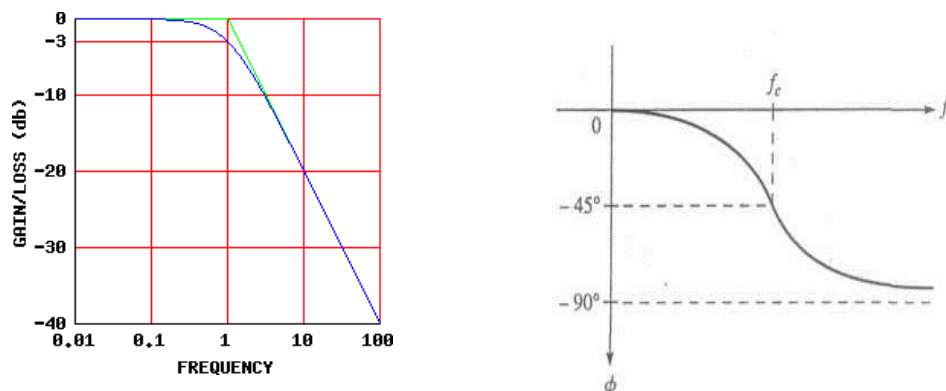


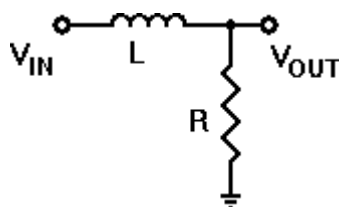
FIGURE: Gain and Phase characteristic of a Low-pass filter

Equipment and Components required:

- Oscilloscope
- Signal Generator
- Inductor
- Resistors(1K, 10K, Use resistor of standard value available in the Lab)

Student Name: _____

Student Reg. & Section: _____

Circuit to be patched on breadboard:**Laboratory Tasks:**

1. Apply a 1 V_{pp} 100 Hz signal as input to the network of LPF and measure the corresponding output voltage level. Determine the decibel gain of the filter.

$$G \text{ (dB)} = 20 \log [V_o/V_{in}]$$

2. Write down the formula for the Cut-off Frequency and phase difference between output voltage and input voltage for RL low pass filter.

3. Determine the phase difference between V_o and V_{in} in degrees from the oscilloscope and calculate it using formula and compare it.

4. Repeat step 1 and 3 for three frequencies lesser than the critical frequency, one at critical frequency and at three frequencies greater than the critical frequency.

Observations & Calculations:Cutoff Frequency: $f_c =$ _____

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	db = 20 log (V_o/V_{in})	$\phi =$ _____ (degrees)	
						Cal	Measured
1	500	1.98	1.86	0.94	-0.54	1.8	1.8
2	1K	1.98	1.86	0.94	-0.54	3.6	7.6
3	10K	1.98	1.58	0.80	-1.94	32.14	32.14
4	15.9K	1.98	1.36	0.69	-3.22	45	45.9
5	20K	1.98	1.22	0.62	-4.15	51.5	51.5
6	50K	1.98	0.60	0.31	-10.2	72.34	72.34

IMPLEMENTATION OF HIGH PASS FILTER:

High Pass Filter: A high pass filter allows signals with higher frequencies to pass from input to output while rejecting lower frequencies. The RL version of the high-pass filter uses a series resistor and a shunt inductor to accomplish its purpose. At high frequencies, X_L is

large, so the inductor is nearly an open circuit for such signals. At low frequencies, X_L is very small, and effectively connects those signals directly to ground. As before, the cutoff frequency occurs where $R = X_L$, so that L is just beginning to have a significant effect on the signal.

Phase Shift in a High-Pass Filter

Both the RC and the RL high-pass filters act as lead networks. The phase shift from input to output for the RL high pass filter is shown in the figure:

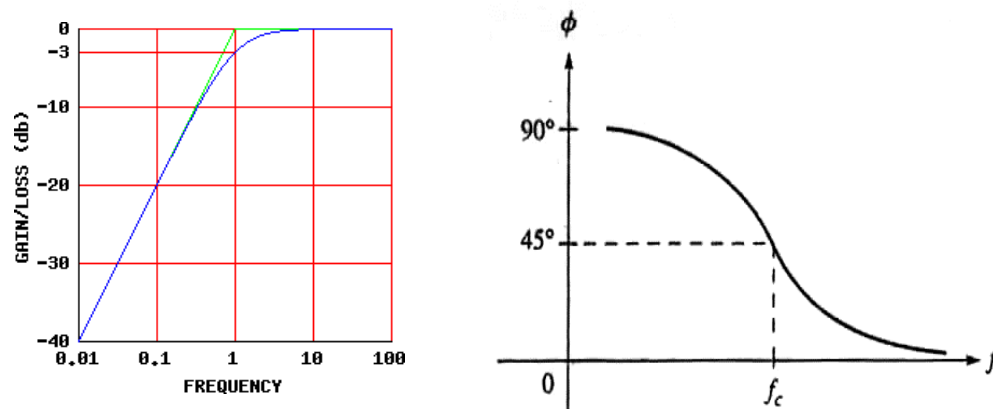
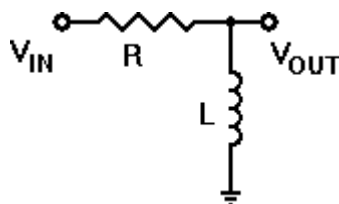


FIGURE: Gain and Phase characteristic of a high-pass filter.

Circuit to be patched on breadboard:



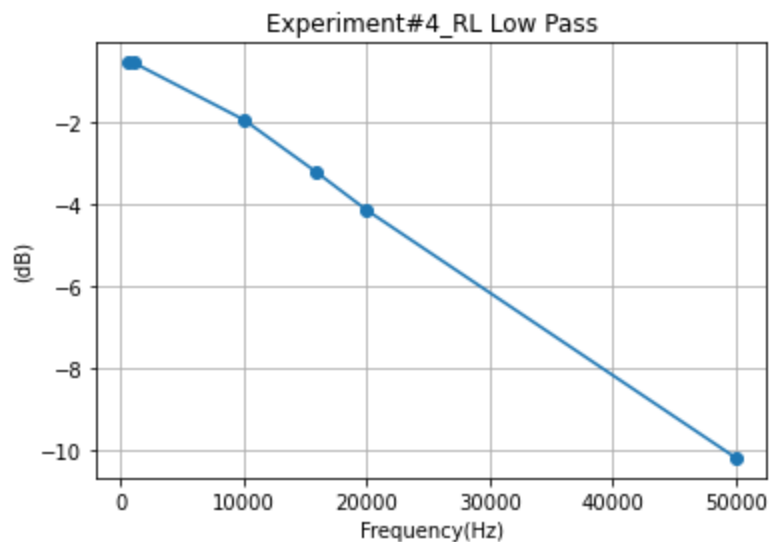
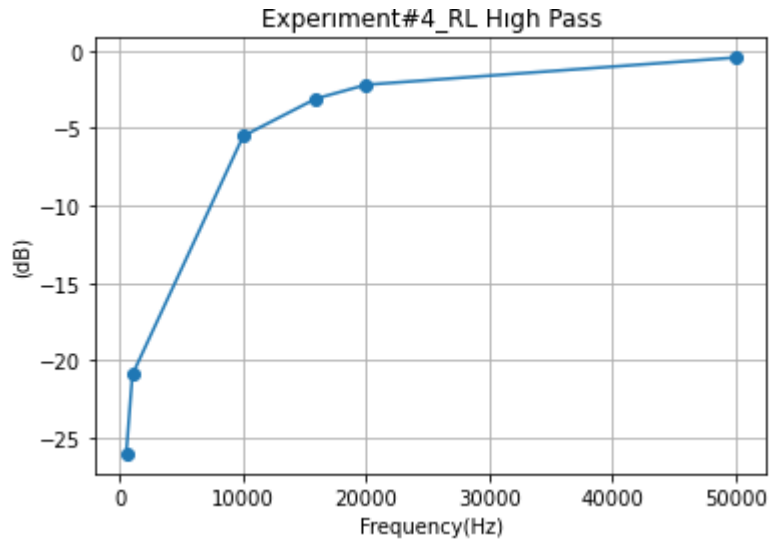
Observations & Calculations:

Cutoff Frequency: $f_c = \underline{15.9K}$

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	db = $20 \log$ (V_o/V_{in})	$\phi =$ _____ (degrees)	
						Cal	Measured
1	500	1.98	0.1	0.05	-26	-88.2	-88.2
2	1K	1.98	0.18	0.09	-20.9	-86.4	-86.4
3	10K	1.98	1.06	0.53	-5.51	-57.86	-57.86
4	15.9K	1.98	1.38	0.7	-3.1	-45	-45.9
5	20K	1.98	1.54	0.78	-2.2	-38.5	-38.5
6	50K	1.98	1.90	0.95	-0.45	-17.66	-17.66

Laboratory Report:

1. Plot the graph between gain against frequency and Phase angle against frequency for low pass filter?
2. Plot the graph between gain against frequency and Phase angle against frequency for high pass filter?
3. Find the cut-off frequency from graphs and compare it with calculated value?



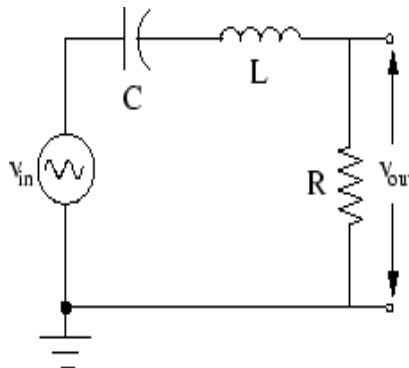
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Experiment 5: To plot the magnitude and phase response of a series resonant band-pass filter.

Equipment and Components required:

- Signal Generator
- Oscilloscope
- Multimeter
- Capacitor: $0.01 \mu\text{F}$ (use capacitor of any value available in the lab)
- Inductor 100-200 mH (use inductor of any value available in the lab)
- Resistors (1/4W): $1\text{K}\Omega$, $2\text{K}\Omega \pm 5\text{ percent}$

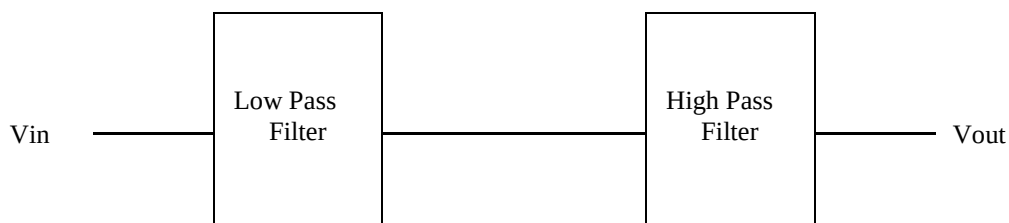
Circuit to be patched on breadboard:



Theory:

Band pass Filter

It allows a certain band of frequencies to pass and attenuates or rejects all frequencies below and above the pass band. A combination of low-pass and high-pass filter can be used to form band pass filters.



Low-Pass and High-Pass Filters used to form a band-pass filter

Operation of Series Resonant Band Pass Filter

A series resonant filter has minimum input impedance. At critical frequency the inductor and the capacitor in series behave as simple resistor. Hence making maximum output across the load resistor.

At the frequencies other than resonant frequencies, the reactance offered by the inductor or capacitor is very large, hence output voltage will be very small at high as well as at low frequencies.

Bandwidth

The bandwidth of a band-pass filter is the range of frequencies for which the current, and therefore the output voltage, is equal or greater than 70.7% of its value at the resonant frequency.

Mathematically, Bandwidth = $\frac{\text{Resonant Frequency } f_r}{\text{Quality Factor } Q}$

Quality factor

Quality Factor is the ratio of reactive power developed in inductor or capacitor to average power dissipated in resistor.

$$\text{Quality Factor} = \frac{\text{Reactive power developed in inductor or capacitor}}{\text{Average power dissipated in resistor}}$$

Quality Factor indicates the selectivity of the filter and can be expressed as,

$$\text{Quality Factor} = \frac{\omega L}{R} = \frac{2\pi f_r L}{R}$$

Laboratory Tasks:

- 1]. For the components used in the circuit, calculate and record the resonant frequencies for the circuit in the fig. Calculate, also, the circuit-Q and bandwidth of the circuit.
- 2]. Construct the circuit shown in fig on the breadboard.
- 3]. At a frequency of 500Hz, adjust V_{in} to some convenient value, such as 5V rms.
- 4]. Use multimeter to measure V_o and record it in table.
- 5]. Vary the frequency, measure and record V_o while maintaining V_{in} constant.
- 6]. Complete the decibel gain row of the table.

Plot the decibel voltage gain ratio versus log frequency.

OBSERVATIONS & CALCULATIONS

Resonant Frequency $f_r = 1 / 2\pi\sqrt{LC}$

Quality Factor $Q = \omega L / R$

Bandwidth = f_r / Q

No.	Input Frequency f (Hz)	Input Voltage Vin (volts)	Output Voltage Vo (volts)	Vo/ Vin (volts)	db = 20 log (Vo/Vin)	$\theta = \tan^{-1}(X_L - X_C/R)$ (degrees)
1	8K	1	0.56	0.56	-5.04	-56
2	10K	1	0.68	0.68	-3.35	-44
3	12K	1	0.78	0.78	-2.16	-29.8
4	14K	1	0.86	0.86	-1.31	-14.42
5	15.9K	1	0.94	0.94	-0.54	-0.1
6	16K	1	0.94	0.94	-0.54	0.6
7	18K	1	0.92	0.92	-0.72	13.9
8	20K	1	0.88	0.88	-1.11	24.7
9	22K	1	0.84	0.84	-1.51	33.4

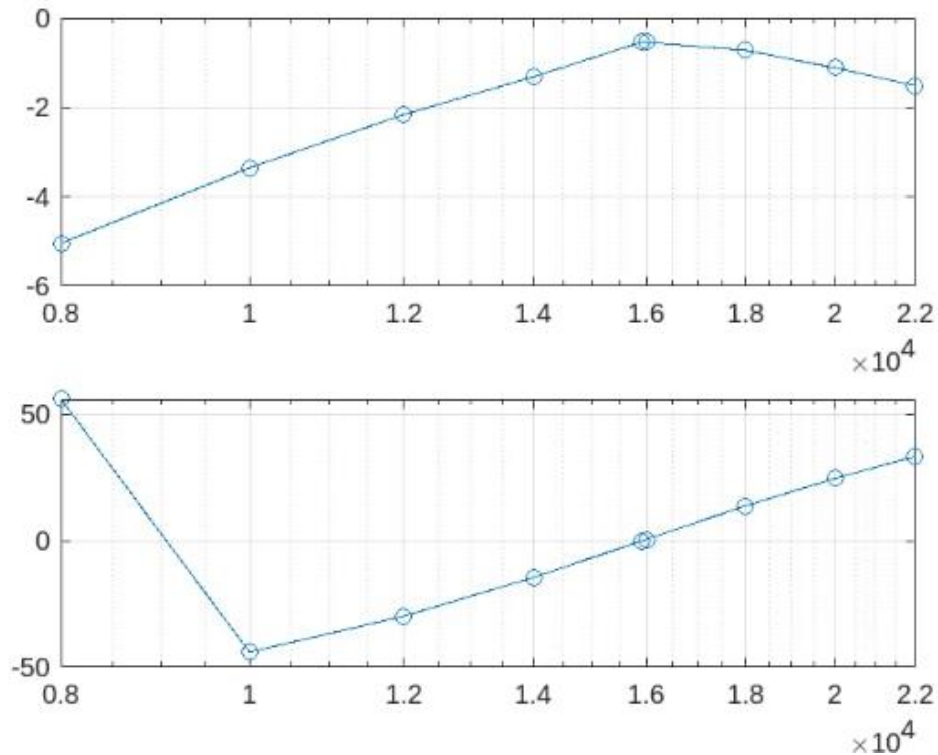
Laboratory Report:

1]. Plot on semilog paper:

I. db against frequency

II. Phase angle against frequency

Magnitude and Phase response of a series band pass filter



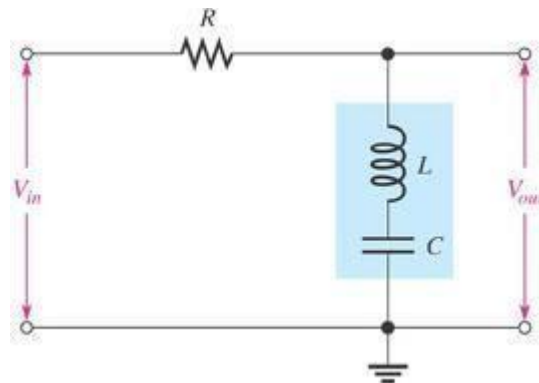
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Experiment 6: To plot the magnitude and phase response of a series resonant band-stop filter.

Equipment and Components required:

- Signal Generator
- Oscilloscope
- Multimeter
- Capacitor: $0.01 \mu F$ (use capacitor of any value available in the lab)
- Inductor 100-200 mH (use inductor of any value available in the lab)
- Resistors (1/4W): $1K\Omega$, $2K\Omega \pm 5$ percent

Circuit to be patched on breadboard:

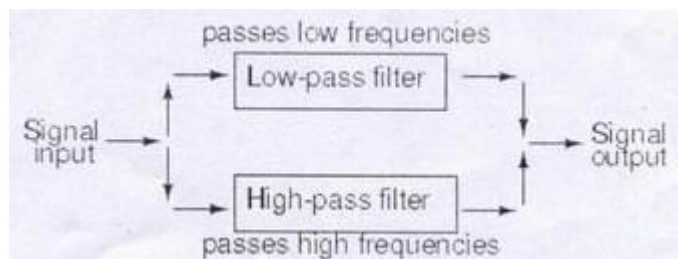


Theory:

Band stop filter

It is a filter that rejects a certain band or range of frequencies while passing all frequencies below and above the rejected band. Band stop filters block signals occurring between two given frequencies, F_L and F_H .

It can be made out of a low-pass and a high pass filter by connecting the two filter sections in parallel with each other instead of in series.



Operation of Bandstop filter

When the series LC combination reaches resonance, its very low impedance shorts out the signal, dropping it across resistor R1 and preventing its passage on to the load. Thus, within the band at which the resonant frequency occurs, there is a relatively less output and that set of frequencies are attenuated.

At frequencies other than resonant frequencies, the reactance offered by inductor and capacitor is very large, thus outside the band at which resonant frequency occurs, there is large output and that set of frequencies are passed to the output.

Corner Frequency

Because a real filter rolls off gradually, you usually specify the corner frequency as the frequency at which the response is $1/\sqrt{2}$ (0.707) of that in the pass band. Because electronic engineers traditionally describe relative signal strengths in decibels, the frequency is also referred to as 3-dB point.

Laboratory Tasks:

- 1]. For the components used in the circuit, calculate and record the resonant frequencies for the circuit in the fig. Calculate, also, the circuit-Q and bandwidth of the circuit.
- 2]. Construct the circuit shown in fig on the breadboard.
- 3]. At a frequency of 500Hz, adjust V_{in} (sinusoidal) to some convenient value, such as 5V rms.
- 4]. Use oscilloscope to measure V_o and record it in table.
- 5]. Vary the frequency, measure and record V_o while maintaining V_{in} constant.
- 6]. Complete the decibel gain row of the table.

Plot the decibel voltage gain ratio versus log frequency.

OBSERVATIONS & CALCULATIONS

Resonant Frequency $f_r = 1 / 2\pi\sqrt{LC}$

Quality Factor $Q = \omega L / R$

Bandwidth = f_r / Q

No.	Input Frequency f (Hz)	Input Voltage Vin (volts)	Output Voltage Vo (volts)	Vo/ Vin (volts)	db = 20 log (Vo/Vin)	$\theta = \tan^{-1}(X_L - X_C/R)$ (degrees)
1	8K	1	0.8	0.8	-1.94	-56
2	10K	1	0.74	0.74	-2.61	-44
3	12K	1	0.56	0.56	-5	-29.8
4	14K	1	0.38	0.38	-8.4	-14.42
5	15.9K	1	0.16	0.16	-16	-0.1
6	16K	1	0.16	0.16	-16	0.6
7	18K	1	0.12	0.12	-18.41	13.9
8	20K	1	0.3	0.3	-10.45	24.7
9	22K	1	0.4	0.4	-8	33.4

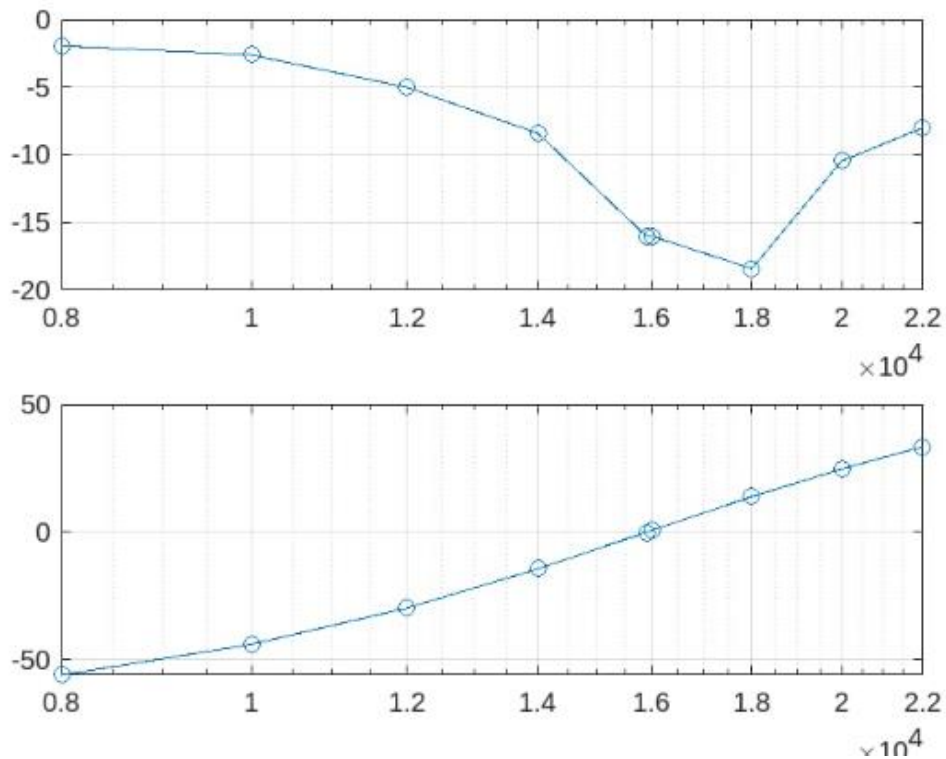
Laboratory Report:

1]. Plot on semilog paper:

I. db against frequency

II. Phase angle against frequency

Magnitude and Phase response of a series band stop filter



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Experiment 7: Implementation of first order Active low pass and high pass filter.

When a filter contains a device like an Op Amp they are called active filters. These active filters differ from passive filters (simple RC circuits) by the fact that there is the ability for gain depending on the configuration of the elements in the circuit. There are some problems encountered in active filters that need to be overcome. The first is that there is still a gain bandwidth limitation that arises. The second is the bandwidth in general. In a high pass filter there is going to be high frequency roll off due to the limitations of the Op Amp used. This is very hard to overcome with conventional op amps. However, active filters have three main advantages over passive filters: Inductors can be avoided. Without them, passive filters cannot obtain a high Q (low damping), but inductors are often large and expensive (at low frequencies), may have significant internal resistance, and may pick up surrounding electromagnetic signals. The shape of the response and the tuned frequency can often be set easily by varying resistors, in some filters one parameter can be adjusted without affecting the others. Variable inductances for low-frequency filters are not practical. The amplifier powering the filter can be used to buffer the filter from the electronic components it drives or is fed from, variations in which could otherwise significantly affect the shape of the frequency response.

IMPLEMENTATION OF LOW PASS FILTER:

The low pass filter is one that allows low frequencies and stops (attenuates) higher frequencies, hence the name. The design of a low pass filter needs to take into consideration the maximum frequency that would need to be allowed through. This is called the cut off frequency (or the 3 dB down frequency). Based on the type of filter that is used (e.g. Butterworth, Bessel, chebyscheff) the attenuation of the higher frequencies can be greater. This attenuation is also based on the order (e.g. 1st, 2nd, 3rd...) of the filter that is used. Based on the order of the filter the roll-off of the filter can be calculated using the formula $-n \times 20 \text{ dB/decade}$. This means that a first order low pass filter has an attenuation of -20 dB/decade , while a second order filter should have -40 dB/decade roll-off and on down the list for higher orders. Shown in Figure 1 is the basic active 1st order low pass filter (in the non-inverting configuration) with unity gain.

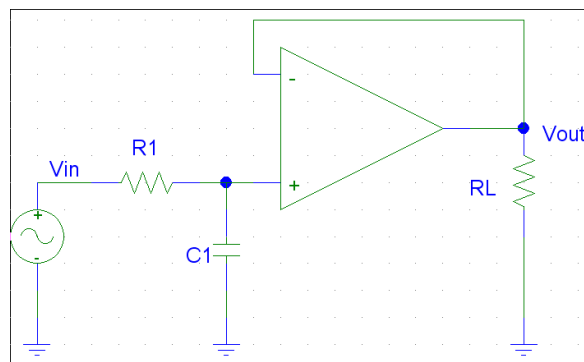


Figure 1: Basic 1st Order Bessel LP Filter in the Non-Inverting Configuration with Unity Gain

The equation in (1) is used to calculate the value of the capacitor needed based on a chosen value for cutoff frequency and R_1 (or vice versa if a value for C_1 and a cutoff frequency are chosen then the value of R_1 can be found). There is unity gain in this configuration because of the non-inverting properties of the Op Amp. To change the gain, the feedback network must be changed to include two other resistors (R_2 and R_3). The gain is then found to be $1 + R_3/R_2$ because of the non-inverting configuration. The circuit with a non-unity gain is shown in Figure 2.

$$f_c = \frac{1}{2\pi R_1 C_1} \quad (1)$$

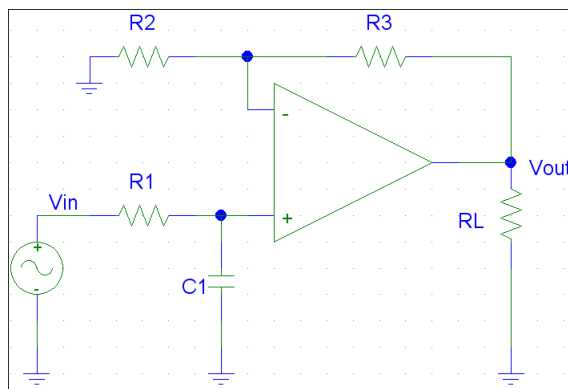


Figure 2: Basic 1st Order Bessel LP Filter in the Non-Inverting Configuration with Non-unity Gain

Equipment and Components required:

- Oscilloscope
- Signal Generator
- Capacitor
- Resistors (Use resistor of standard value available in the Lab)

Circuit to be patched on breadboard:

- Patch both the above circuits and verify the output of Low pass filter with unity and no unity gain.

Laboratory Tasks:

1. Choose appropriate value of R_1 and C_1 . $R_1 = 70\text{mV} / 500\text{nA}$, so R_1 comes out to be 140kohms . You can use standard value of 120kohms . Similarly use appropriate value of C_1 (0.001micro) so that cut-off frequency is not very small. For the filter with non-unity gain set the Gain=5 and then implement it.

2. Apply a 1 Vpp 100 Hz signal as input to the network of LPF and measure the corresponding output voltage level. Determine the decibel gain of the filter.

$$G (\text{dB}) = 20 \log [V_o/V_{in}]$$

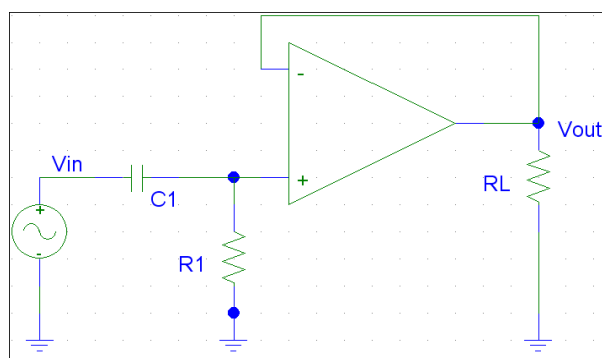
3. Repeat step 1 and 2 for three frequencies lesser than the critical frequency, one at critical frequency and at three frequencies greater than the critical frequency.

Observations & Calculations:**Cutoff Frequency: $f_c = 15.9K$**

No.	Input Frequency f (Hz)	Input Voltage Vin (volts)	Output Voltage Vo (volts)	Vo/ Vin (volts)	db = 20 log (Vo/Vin)
1	500	2	2	1	0
2	1K	2	2	1	0
3	5K	2	1.96	0.98	-0.175
4	10K	2	1.72	0.86	-1.31
5	15.9K	2	1.44	0.72	-2.85
6	20K	2	1.16	0.58	-4.73
7	25K	2	1.10	0.55	-5.19
8	30K	2	0.96	0.42	-6.38

IMPLEMENTATION OF HIGH PASS FILTER:

High Pass Filter: A high pass filter allows signals with higher frequencies to pass from input to output while rejecting lower frequencies. If creating a low pass filter was easy, then creating a high pass filter is even easier. In the case of the 1st order Bessel LP filter the capacitor and resistor only need to be interchanged with each other and the result is a high pass filter. The same equation holds for finding the cutoff frequency and is shown in (1). The circuit shown in Figure 4 is that of a 1st order Bessel HP filter with unity gain. The gain can be adjusted to the non-unity case by adding the feedback network resistors in the same location as the LP circuit of Figure 2.

Circuit to be patched on breadboard:**Figure 3: 1st Order Bessel HP Filter with unity gain**

Student Name: _____

Student Reg. & Section: _____

Observations & Calculations:

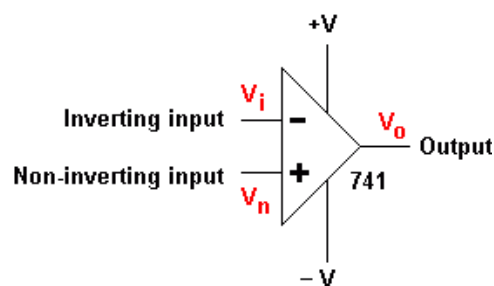
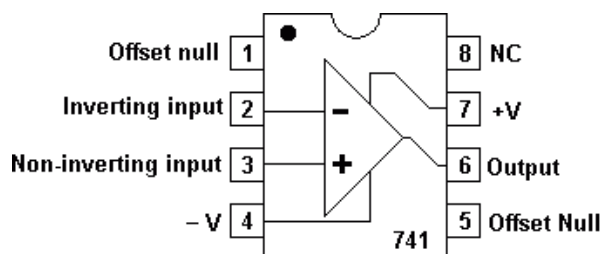
(Choose $C_1 = 0.001$ micro Farads and find the value of R_1 for any value of f_c (say $f_c = 5$ kHz))

Cut off Frequency: $f_c = 15.9K$

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	db = $20 \log (V_o/V_{in})$
1	1K	2	0.138	0.06	-23.2
2	5K	2	0.56	0.28	-11.1
3	10K	2	0.96	0.48	-6.38
4	15.9K	2	1.32	0.66	-3.61
5	20K	2	1.46	0.73	-2.73
6	25K	2	1.56	0.78	-2.16
7	30K	2	1.64	0.82	-1.72
8	40K	2	1.74	0.87	-1.21

Pin Layout of OP-AMP:

Biasing Voltage = +15V

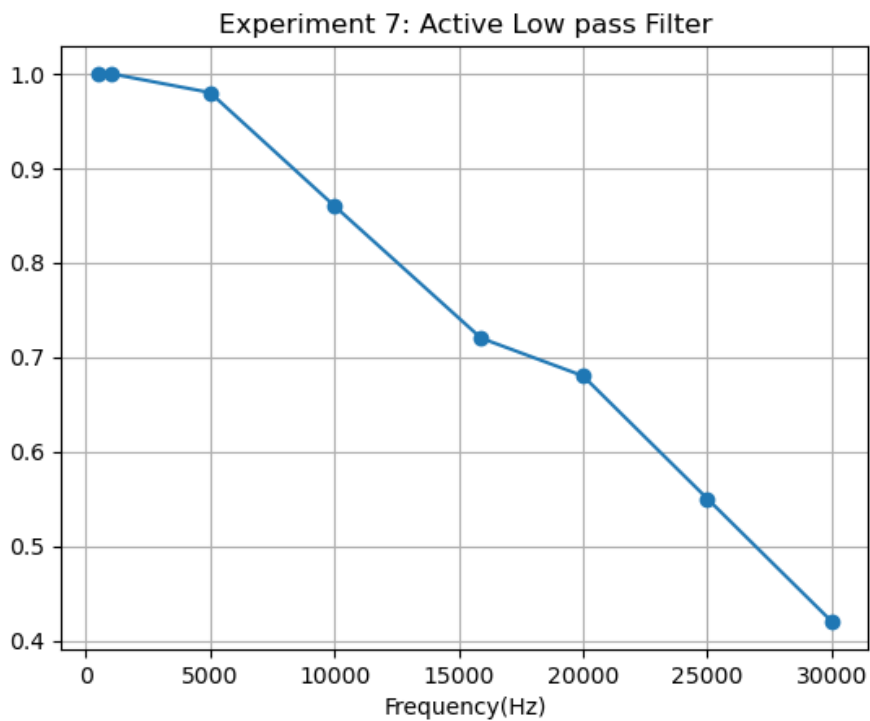
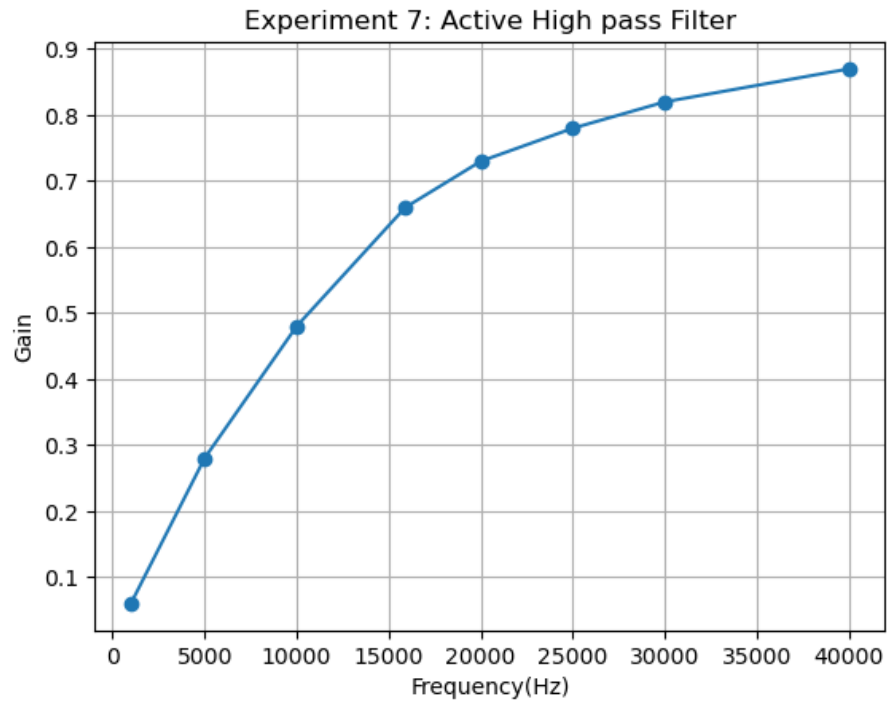


Laboratory Report:

1. Plot the graph between gain against frequency for low pass filters?
2. What is the gain of the First Order LP Filter in the pass band? How does this compare to the theoretical value? What is the roll off rate of the filter in the stop band?
3. Plot the graph between gain against frequency for high pass filter?
4. Find the cut-off frequency from graphs and compare it with calculated value?

Student Name: _____

Student Reg. & Section: _____



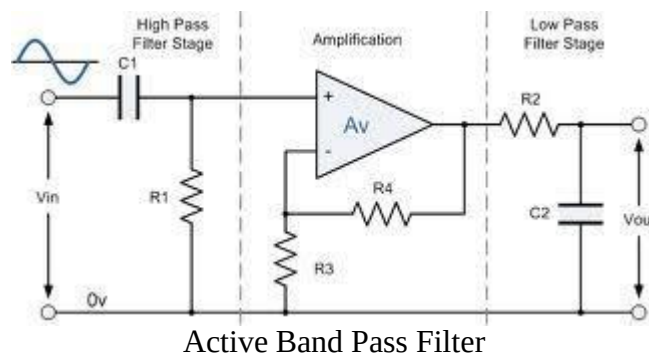
University of Engineering and Technology
Department of Electrical Engineering
EE 110 Circuit Analysis and Design Laboratory

Experiment 8: To plot the magnitude and phase response of a Active Band Pass and Band Stop filters.

Equipment and Components required:

- Signal Generator
- Oscilloscope
- Multimeter
- OP-AMPS
- Capacitor: $0.01 \mu\text{F}$ (use capacitor of any value available in the lab)
- Inductor 100-200 mH (use inductor of any value available in the lab)
- Resistors (1/4W): $1\text{K}\Omega$, $2\text{K}\Omega \pm 5$ percent

Circuit to be patched on breadboard:



Theory:

Active Band Pass Filter is a frequency selective filter circuit used in electronic systems to separate a signal at one particular frequency, or a range of signals that lie within a certain "band" of frequencies from signals at all other frequencies. This band or range of frequencies is set between two cut-off or corner frequency points labeled the "lower frequency" (f_L) and the "higher frequency" (f_H) while attenuating any signals outside of these two points.

Simple Active Band Pass Filter can be easily made by cascading together a single Low Pass Filter with a single High Pass Filter as shown.

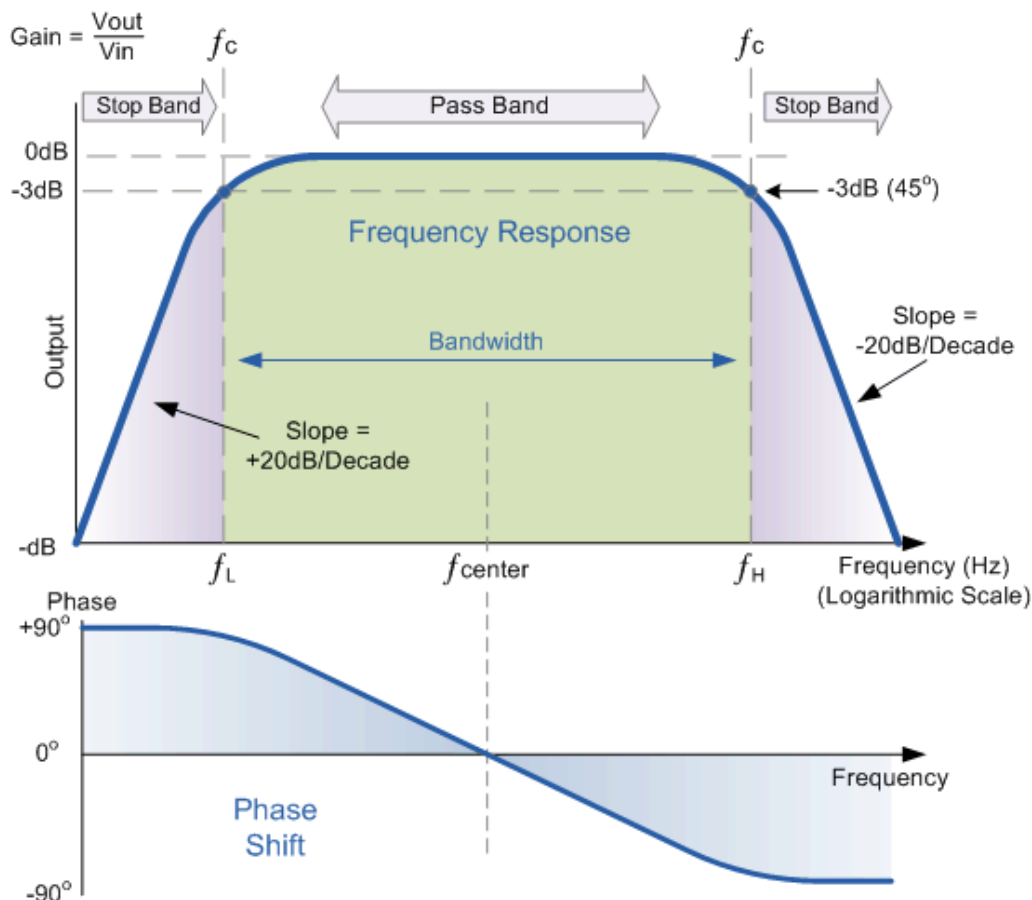


The cut-off or corner frequency of the low pass filter (LPF) is set higher than the cut-off frequency of the high pass filter (HPF) and the difference between the frequencies at the -3dB point will determine the "bandwidth" of the band pass filter while attenuating any signals outside of these points. One way of making a very simple Active Band Pass Filter is to connect the basic passive high and low pass filters we look at previously to an amplifying op-amp circuit as shown in the figure above.

This cascading together of the individual low and high pass passive filters produces a low "Q-factor" type filter circuit which has a wide pass band. The first stage of the filter will be the high pass stage that uses the capacitor to block any DC biasing from the source. This design has the advantage of producing a relatively flat asymmetrical pass band frequency response with one half representing the low pass response and the other half representing high pass response.

The higher corner point (f_H) as well as the lower corner frequency cut-off point (f_L) are calculated the same as before in the standard first-order low and high pass filter circuits. Obviously, a reasonable separation is required between the two cut-off points to prevent any interaction between the low pass and high pass stages. The amplifier provides isolation between the two stages and defines the overall voltage gain of the circuit. The bandwidth of the filter is therefore the difference between these upper and lower -3dB points.

The normalized frequency response and phase shift for an active band pass filter will be as follows.



Laboratory Tasks:

- 1]. From the chosen components, calculate and record the lower cut off frequency, resonant frequency and higher cut-off frequency for the circuit in the fig. Also calculate the bandwidth of the circuit.
- 2]. Construct the circuit shown in fig on the breadboard.
- 3]. At a frequency of 500Hz, adjust V_{in} (sinosidal) to some convenient value, such as 5V rms.
- 4]. Use multimeter/oscilloscope to measure V_o and record it in table.
- 5]. Vary the frequency to measure and record V_o while maintaining V_{in} constant at nine different values of frequency including lower, center and higher cut-off frequency.
- 6]. Complete the decibel gain row of the table.
- 7]. Plot the decibel voltage gain ratio versus log frequency.
- 8]. Try to modify the circuit or change the component values to construct BANDSTOP filter.

OBSERVATIONS & CALCULATIONS

Lower cut-off Frequency =
Higher cut-off Frequency =
Center Frequency =
Bandwidth =

FOR BANDPASS FILTER:

No.	Input Frequency f (Hz)	Input Voltage V_{in} (volts)	Output Voltage V_o (volts)	V_o/V_{in} (volts)	db = 20 log (V_o/V_{in})	θ = (degrees)
1	4K	9.5	1.4	0.14	-17	
2	6K	9.5	1.5	0.15	-16.4	
3	7K	9.5	1.8	0.18	-14.9	
4	10.73K	9.5	1.5	6.667	-16.5	-89
5	14K	9.5	1.2	0.12	-18.4	
6	16K	9.5	1.1	0.11	-19.2	
7	18K	9.5	1.0	0.1	-20	

Lower cut-off Frequency =
 Higher cut-off Frequency =
 Center Frequency =
 Stop band =

FOR BANDSTOP FILTER:

No.	Input Frequency f (Hz)	Input Voltage Vin (volts)	Output Voltage Vo (volts)	Vo/ Vin (volts)	db = 20 log (Vo/Vin)	θ = (degrees)

Laboratory Report:

1]. Plot on semilog paper:

I. db against frequency

II. Phase angle against frequency

Magnitude and Phase response of a Active band pass filter

